

Operating Systems – Lecture Review Questions

1. According to the lecture, what is the main purpose of an operating system?

- A) To manage hardware and provide an environment for executing programs conveniently and efficiently
- B) To compile and run user programs without any hardware interaction
- C) To act solely as a resource allocator for I/O devices
- D) To focus only on user interface design

Answer: A) To manage hardware and provide an environment for executing programs conveniently and efficiently

(The lecture defines OS as a program that manages hardware and provides a convenient, efficient execution environment.)

2. From the user view, what is the primary design focus of an operating system?

- A) Resource utilization
- B) Ease of use, with some attention to performance
- C) Maximizing CPU time allocation
- D) Minimizing memory space usage

Answer: B) Ease of use, with some attention to performance

(The user view emphasizes ease of use and performance, with no focus on resource utilization.)

3. From the system view, how is an operating system described?

- A) As a user interface tool
- B) As a resource allocator and manager
- C) As a program compiler
- D) As a hardware simulator

Answer: B) As a resource allocator and manager

(The system view sees the OS as managing resources like CPU time, memory, and I/O devices.)

4. What is the role of the bootstrap program in a computer system?

- A) To manage user processes after boot
- B) To initialize the system and load the operating system when powered up
- C) To handle interrupts during runtime
- D) To allocate memory for applications

Answer: B) To initialize the system and load the operating system when powered up

(The bootstrap program is stored in ROM/EEPROM and initializes CPU registers, devices, and loads the OS.)

5. How is an interrupt typically triggered in a computer system?

- A) By user commands only
- B) By hardware signals or software system calls
- C) By memory allocation requests
- D) By file system operations

Answer: B) By hardware signals or software system calls

(Interrupts can come from hardware via the system bus or software via system calls.)

6. What happens when the CPU receives an interrupt?

- A) It continues executing the current task
- B) It stops the current task and transfers execution to a fixed location for the service routine
- C) It reboots the system
- D) It allocates more memory

Answer: B) It stops the current task and transfers execution to a fixed location for the service routine

(The CPU halts current execution and jumps to the interrupt service routine address.)

7. What is the function of the interrupt vector?

- A) To store user programs
- B) To index addresses of interrupt service routines by device number
- C) To manage memory allocation
- D) To handle file backups

Answer: B) To index addresses of interrupt service routines by device number

(The interrupt vector provides the address of the service routine based on the interrupting device's unique number.)

8. What does DMA stand for in the context of operating systems?

- A) Direct Memory Allocation
- B) Direct Memory Access
- C) Device Management Algorithm
- D) Data Migration Application

Answer: B) Direct Memory Access

(DMA is mentioned as Direct Memory Access in the lecture.)

9. In dual-mode operation, what does the mode bit indicate?

- A) User mode (0) or kernel mode (1)
- B) Kernel mode (0) or user mode (1)
- C) Memory allocation status
- D) Interrupt priority level

Answer: B) Kernel mode (0) or user mode (1)

(The mode bit distinguishes kernel (0) from user (1) mode.)

10. What happens if a privileged instruction is attempted in user mode?

- A) It executes normally
- B) The hardware treats it as illegal and traps it to the OS
- C) The system reboots
- D) It switches to multimode

Answer: B) The hardware treats it as illegal and traps it to the OS

(Privileged instructions are only allowed in kernel mode; user mode attempts are trapped.)

11. In process management, what is a key distinction between a program and a process?

- A) A program is active, while a process is passive
- B) A program is passive (like file contents), while a process is active
- C) Both are the same entity
- D) Processes are stored on disk only

Answer: B) A program is passive (like file contents), while a process is active

(The lecture emphasizes that a program is passive, but a process is an active entity.)

12. Which activity is NOT part of OS process management?

- A) Scheduling processes on CPUs
- B) Providing process synchronization mechanisms
- C) Allocating disk space for files
- D) Creating and deleting processes

Answer: C) Allocating disk space for files

(Disk space allocation is under storage management; the others are process management activities.)

13. What is main memory described as in memory management?

- A) A small array of bits
- B) A large array of bytes with individual addresses
- C) A repository for I/O devices only
- D) A slow-access storage device

Answer: B) A large array of bytes with individual addresses

(Main memory is a large array of bytes, shared by CPU and I/O, with each byte addressed.)

14. In file-system management, what is NOT an OS responsibility?

- A) Creating and deleting files
- B) Mapping files to secondary storage
- C) Scheduling CPU threads
- D) Backing up files on nonvolatile media

Answer: C) Scheduling CPU threads

(CPU scheduling is process management; the others are file-system activities.)

15. What is a key activity in mass-storage management?

- A) Process synchronization
- B) Disk scheduling and free-space management
- C) User interface design
- D) Interrupt vector indexing

Answer: B) Disk scheduling and free-space management

(Mass-storage involves free-space management, allocation, and disk scheduling for efficiency.)

16. What is caching in the context of storage hierarchy?

- A) Permanent data storage
 - B) Temporary storage of frequently used data in faster media
 - C) Deleting unused files
 - D) Allocating CPU time
- Answer:** B) Temporary storage of frequently used data in faster media
(Caching is implied in the storage hierarchy for faster access.)
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17. What issue does cache coherency address?

- A) Data duplication in hierarchical storage levels
 - B) Interrupt handling
 - C) Process creation
 - D) File backup
- Answer:** A) Data duplication in hierarchical storage levels
(Cache coherency ensures consistency when data appears in multiple levels, like disk, memory, and cache.)
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18. What components make up the I/O subsystem?

- A) Only device drivers
 - B) Device-driver interface, specific drivers, and memory management (buffering, caching, spooling)
 - C) Bootstrap programs
 - D) Virtualization layers
- Answer:** B) Device-driver interface, specific drivers, and memory management (buffering, caching, spooling)
(The I/O subsystem includes these components for handling I/O operations.)
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19. What is the purpose of protection and security in operating systems?

- A) To manage cloud services
 - B) To safeguard resources and data
 - C) To emulate hardware
 - D) To schedule disks
- Answer:** B) To safeguard resources and data
(Though brief, protection and security are listed as key OS aspects.)
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20. What is virtualization in operating systems?

- A) Running OS as applications within other OS
- B) Physical hardware replacement
- C) File system mapping
- D) Interrupt simulation

Answer: A) Running OS as applications within other OS

(Virtualization allows OS to run as apps on other OS, enabling platform portability.)

21. What is emulation used for in virtualization?

- A) When source and target CPU types differ
- B) For identical CPU types only
- C) To speed up native code
- D) For cloud backups

Answer: A) When source and target CPU types differ

(Emulation, like Apple's Rosetta, handles different CPU types but at a performance cost.)

22. Which is NOT a type of cloud computing mentioned?

- A) Public cloud
- B) Private cloud
- C) Hardware as a Service (HaaS)
- D) Hybrid cloud

Answer: C) Hardware as a Service (HaaS)

(Types include public, private, hybrid, SaaS, PaaS, IaaS; HaaS is not listed.)

23. What is Software as a Service (SaaS) in cloud computing?

- A) Servers for storage
- B) Applications like word processors available via the Internet
- C) A software stack for databases
- D) Hybrid cloud components

Answer: B) Applications like word processors available via the Internet

(SaaS provides applications over the Internet, e.g., word processors or spreadsheets.)

24. Where are embedded systems commonly found?

- A) Only in desktops
- B) In car engines, robots, DVDs, and microwave ovens

C) Exclusively in cloud servers

D) In open-source kernels

Answer: B) In car engines, robots, DVDs, and microwave ovens

(Embedded systems are prevalent in everyday devices like these.)

25. What characterizes real-time operating systems in embedded systems?

A) Flexible time constraints

B) Well-defined, fixed time constraints where processing must meet deadlines

C) No time requirements

D) Focus on user ease only

Answer: B) Well-defined, fixed time constraints where processing must meet deadlines

(Real-time systems have rigid time requirements; failure occurs if constraints are missed.)

26. Which is an example of an open-source operating system?

A) Microsoft Windows

B) Linux

C) Mac OS X (fully)

D) iOS

Answer: B) Linux

(Linux is famous for being open-source, available in source-code format.)

27. What approach does Microsoft Windows take regarding source code?

A) Fully open-source

B) Closed-source (compiled binary only)

C) Hybrid like Mac OS X

D) Emulation-based

Answer: B) Closed-source (compiled binary only)

(Windows is a closed-source example, unlike open-source Linux or hybrid Mac OS X.)

Wish you all the best!

Fathy.